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$$a(2iz - 3 + i) + b(z + 1 + i) = 5$$

$$\Rightarrow 2iaz - 3a + ai + bz + b + bi = 5$$

$$\Rightarrow \begin{cases} (2i a + b) z & = 0, \\ -3a + ai + b + bi & = 5 \end{cases}$$

$$\Rightarrow \begin{cases} b & = -2i a, \\ -a - a i & = 5 \end{cases}$$

$$\Rightarrow \begin{cases} a & = -\frac{5}{1+i} = -\frac{5(1-i)}{2}, \\ b & = -2i \left(-\frac{5}{1+i} \right) = 5 + 5i \end{cases}$$

References